Analysis of Variance (ANOVA)

Def.: Anova is a statistical method used to compare the means of 2 or more groups.

ANOVA:

1. Factors (Variables)
2. Levels

Ex, Medicine (Factor)

[Dosage] 5 mg 10 mg 15 mg → Levels

Ex, Mode of Payment (Factor)

[Levels] G-pay Phone pe IMPS NEFT

⇒ Assumptions In Annova :

① Normality of sampling distribution of mean.
The distribution of sample mean is normally distributed.

② Absence of Outlier.
Outlying score need to be removed from dataset.

③ Homoginity of Variance.
Population Variance in different level of each independent Variable [Factors] are equal.

$$[\sigma_1^2 = \sigma_2^2 = \sigma_3^2]$$

④ Sample are independent and Randomly chosen.

⇒ Type of Annova :
(3 types)

1. One way Annova : One factor with atleast 2 levels, these levels are independent.

Eg

Doctor wants to test a new medication to decrease a Headache. They split the participants in 3 Condition [10 mg, 20 mg, 30 mg]. Doctor ask the participantes to rate the headache [1 - 10] on a basis.

Ans:

Medication → factor

levels →	10 mg	20 mg	30 mg
	5	7	2
	3	4	6
	-	-	-
	-	-	-

② <u>Repeated</u> <u>Measure</u> <u>Anova</u> : One factor with atleast 2 levels, <u>levels are dependent.</u>

Running → factor

levels →	Day 1	Day 2	Day 3
	8	5	4
	7	4	9
	-	-	-
	-	-	-

③ <u>Factorial</u> <u>Annova</u> : Two or more factors each of which with atleast 2 levels, levels can be independent or dependent.

$\boxed{\text{Running} \rightarrow \text{factor}}$

	level ↓	Day 1	Day 2	Day 3
Gender → Factor	Male	8	5	4
		9	4	3
	Female	2	4	6
		7	8	3

⇒ Analysis of Variance :
(ANOVA)

- Hypothesis Testing In ANOVA
(Partitioning of Variance in the Anova)

1. — ~~Variance~~ S.D. is spread btw two element.
— ~~S.D.~~ Variance is how much far from the mean of each data

→

Null Hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 \cdots\cdots \mu_K$

Alternate Hypothesis H_1 : Atleast one of the sample mean is not equal
$\mu_1 \neq \mu_2 \neq \mu_3 \cdots \mu_K$

→ Test statistics : [F - test]

$$F = \frac{\text{Variance between sample}}{\text{Variance within sample}}$$

Variance between samples [for features]

Variance within Sample

x_1	x_2	x_3
1	6	5
2	7	6
4	3	3
5	2	2
3	1	4
$\bar{x}_1 = 3$	$\bar{x}_2 = 19/5$	$\bar{x}_3 = 4$

H_0 $\bar{x}_1 = \bar{x}_2 = \bar{x}_3$

H_1 Atleast one sample mean is not equal.

One way Annova

One Factor with atleast 2 levels, levels are independent

(1) Doctors wants to test a new medication which reduces headache. They splits the participant into 3 condition [15 mg, 30mg, 45 mg]. Later on the doctor ask the patient to rate the headache between [1 - 10]. Are there any differences btw the 3 conditions alpha = 0.05 ?

Ans :

15 mg	30mg	45 mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

(1) Define Null & Alternate Hypothesis :

H_0 : $\mu_{15} = \mu_{30} = \mu_{45}$

H_1 : not all μ are equal.

(2) Significance $\alpha = 0.05$ C.I. = 0.95

(3) Calculation Degree of freedom.

$N = 21$ $a = 3$ $n = 7$
(7 x 3) (features) (rows)
r x c

$\therefore df_{\text{between samples}} = a - 1 = 3 - 1 = 2$

$\therefore df_{\text{within sample}} = N - a = 21 - 3 = 18$

$\therefore df_{\text{total}} = N - 1 = \underline{\underline{20}}$

$df_1 \quad df_2$

$(2, 18)$

⇓

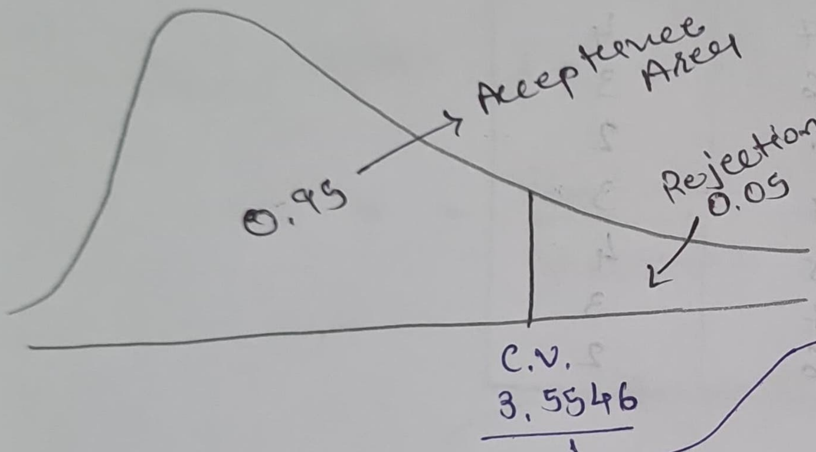
F - table

$\alpha = 0.05$

⇓

find the Critical Value.

(4) Decision Boundary.



from F-table using value of α, df-1, df-2

Decision Rule:

- If F is greater than 3.5546, Reject H_0

otherwise

~~Accept H_0~~ fail to reject H_0

(5) Calculate F-statistics:

$$F = \frac{\text{Variance between Sample}}{\text{Variance within Sample}}$$

	S.S. [Sum of Square]	d.f. [degree freedom]	M.S. [Mean Square]	F.
Between	98.67	2	49.34	
Within	10.29	18	0.54	
Total	108.96	20	49.88	

1.// S.S. between $= \frac{\sum(\sum a_i)^2}{n} - \frac{T^2}{N}$

15 mg : 9 + 8 + 7 + 8 + 8 + 9 + 8 = 57
30 mg : 7 + 6 + 6 + 7 + 8 + 7 + 6 = 47
45 mg : 4 + 3 + 2 + 3 + 4 + 3 + 2 = 21

$= \frac{57^2 + 47^2 + 21^2}{7} - \frac{[57^2 + 47^2 + 21^2]}{21}$

$= \underline{98.67}$

2.// S.S. within $= \sum y^2 - \frac{\sum(\sum a_i)^2}{n}$

$\sum y^2 = 9^2 + 8^2 + 7^2 + 8^2 + 8^2 \ldots$
$= 853$
$= 853 - \frac{[57^2 + 47^2 + 21^2]}{7}$

$= \underline{10.29}$

$$F_{test} = \frac{MS_{between}}{MS_{within}}$$

$$= \frac{49.34}{0.54} = \underline{\underline{86.56}}$$

$\therefore$ If F is greater than 3.5546, Reject H_0

$86.56 > 3.5546$ Reject H_0